

Hub Cultural: An Instagram-Mined, Human-Curated Pipeline for Discovering Cultural Events in France

From a Diaspora Network to a General Cultural Discovery Platform

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Abstract

Culturally-relevant activity in the Paris region — with priority given to photography, cinema, literature, theatre, and locally-organized community events — is rarely centralized: it lives across hundreds of Instagram accounts, is announced inconsistently, and is legible mainly to whoever already follows the right accounts. This project, which originated from but is no longer limited to the specific case of the Latin American diaspora in France, builds a pipeline that scrapes Instagram profiles and posts into a Neo4j graph, applies a three-layer NLP/LLM event-extraction pipeline validated by three rounds of blind human-versus-pipeline evaluation (agreement rising from 58% to 96.7%, precision from 56% to 97.3%), and treats hand curation of accounts — rather than an automated classifier, which under 10% of the time selected a culturally valuable account — as the operative trust signal. A research-operations discipline (a real-time decision log, a run log, and dry-run-first iteration on a continuously growing, shared database) is argued to be part of the methodology rather than incidental project management. The pipeline feeds a public, bilingual, static discovery website with no backend, no user accounts and no personal data collection, whose publication cycle — staged human review of every detected event, a rolling retention window, and a deliberate, logged manual deploy step — is itself part of what this thesis reports as method. A byproduct of the graph, exact PageRank and betweenness centrality over the account network, still feeds one term of the site’s ranking formula and one badge, but is reported here only to that extent: the account-selection role it was originally built for was superseded by human curation, and its fuller exploitation is left as future work. The thesis’s contribution is threefold: methodological (human curation as an evidenced alternative to automated account scoring; a blind evaluation protocol that repeatedly and measurably drove pipeline redesign; a transparent account of where automated signals fail silently); empirical (a mapped, if partial, cultural network with a documented, improving detection accuracy); and applied (a live tool a member of the community it describes can actually use). The project’s scope itself evolved, from a nationality-based inclusion criterion to a purely cultural one, after a blind evaluation showed the earlier criterion was not only a conceptual boundary but a measurable bias baked into the detection pipeline’s own reference vocabulary.

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Chapter 1

Introduction

1.1 Context and Motivation

Hub Cultural began as an attempt to make visible something that is well known anecdotally but poorly documented empirically: the cultural life that Latin American, and in particular Colombian, communities sustain in the Paris region through small associations, independent venues, cultural attachés, and a dense but scattered presence on Instagram. The author’s own Colombian background supplied both the initial motivation and the first seed account for data collection — the Colombian consulate in Paris (@consuladocolparis) — from which the network was grown outward through automatic relationship discovery.

As the project progressed, the boundary that had organized it from the start — *is this account or event part of the Latin American diaspora?* — proved both practically brittle and conceptually narrower than what the data, and the underlying research interest, actually supported. Many of the best-connected cultural accounts discovered through the network turned out to be French institutions, mixed-nationality collectives, or venues with no diaspora affiliation at all, but which were unmistakably part of the same cultural ecosystem the project set out to map — programming photography exhibitions, independent cinema, literary events, and theatre alongside explicitly Colombian or Latin American programming. The project’s scope therefore evolved in two steps: first from Colombia-specific to Latin-American-diaspora-wide, and subsequently from a nationality-based criterion to a purely cultural one. The operative question became not *who is behind this account* but *is this culturally relevant*, with explicit priority — inherited from the project’s original four thematic agendas (`docs/objetivos.md`) — given to photography, cinema, literature, and theatre, alongside a fifth, less formally bounded category: events organized by and for local communities, regardless of national or cultural origin.

This evolution is treated in this thesis not as scope creep to be apologized for, but as a methodological finding in its own right (§6.1): a nationality-based inclusion criterion

turned out to be both harder to operationalize consistently and less faithful to how cultural networks actually cohere in practice, where diaspora and host-society cultural production overlap and reinforce each other rather than remaining cleanly separable. It also turned out to be measurably encoded in the pipeline itself: the semantic reference set used by the first stage of event detection was written entirely in the vocabulary of Colombian diaspora events, and a blind evaluation of 100 posts showed that non-diaspora cultural accounts consequently lost real events at roughly three times the rate of diaspora accounts (§3.8). The scope question was therefore never purely conceptual; it had a direct, quantifiable effect on what the system was able to see.

1.2 Problem Statement and Research Question

The cultural activity produced by, and adjacent to, migrant and diaspora communities in a host country is rarely centralized: it lives across dozens of Instagram accounts, is announced inconsistently, and is legible mainly to whoever already follows the right accounts. This is true of Latin American cultural life in the Paris region specifically, but — as this project’s own data eventually made clear — it is not a problem unique to any one national community; it is a structural feature of how small-scale cultural production circulates on social media generally.

This thesis asks: *How is culturally-relevant activity — with priority given to photography, cinema, literature, theatre, and local community cultural life — organized as a network of accounts and events in the Paris region, and can that structure be systematically mapped, analyzed, and made discoverable using publicly available Instagram data?*

This breaks into three sub-questions:

1. **Structural.** Who are the connective actors in this network — which accounts function as hubs or bridges between otherwise separate cultural circles, and does this differ between institutional accounts and individual or grassroots ones?
2. **Thematic.** How is cultural activity distributed across the project’s priority domains (photography, cinema, literature, theatre, community events), and how well does an automated pipeline classify it into these domains compared to human judgment?
3. **Practical.** Can the resulting graph be turned into a public-facing tool that helps the community it describes actually discover and attend these events — and what does building that tool reveal about the reliability of the underlying data?

1.3 Objectives

The project’s objectives, as originally framed (`docs/objetivos.md`) and updated here to reflect the current, cultural-relevance-based scope:

1. Build and maintain four thematic monitoring agendas — *literature*, *photography*, *cinema*, and *theatre* — tracking the organizations that play an active role in each, complemented by a fifth, less formally bounded priority: cultural events organized by and for local communities.
2. Consolidate a pipeline that extracts events from the accounts under these agendas, primarily from Instagram and, prospectively, from websites and other social networks.
3. Map the network of actors within each thematic agenda and categorize them — by type of organization, geographic zone, cultural identity where relevant, and structural role in the network.
4. Evaluate the participation opportunities these networks open up, using a deliberately human, non-technical success criterion: on this dimension, the project succeeds when it leads to actually meeting someone through it.

Geographically the project centers on Île-de-France, though a meaningful share of curated accounts and events originate from, or are organized by, institutions physically located outside France (§3.14). Instagram is the sole data source, a deliberate boundary rather than an oversight: it is the platform where this cultural activity is, in practice, announced, and only public accounts are in scope. Inclusion is governed by cultural relevance rather than national origin, with the four thematic pillars and the community-events category acting as explicit priorities rather than hard filters.

Chapter 2

Literature Review / Theoretical Framework

This chapter frames the project along two axes that the project’s own history forces it to hold together. The first is the diaspora-studies axis, which motivated the work and still describes a substantial part of the corpus. The second is the cultural-sector axis: since the operative inclusion criterion is now cultural relevance rather than national origin (§6.1), the theoretical framing has to account for the network as a field of cultural production in a metropolitan area, not only as a diaspora formation. The remaining sections supply the methodological grounding — social media as data, graph databases, and NLP event extraction — that Chapters 3 and 4 draw on. Social network analysis proper is treated only briefly (§2.3), since it is no longer this project’s active method (§3.5).

2.1 Diaspora Studies and Cultural Networks

[TODO: Situate within diaspora/migration studies: how migrant communities organize culturally, the role of associations and informal networks, prior work on Latin American diasporas in Europe specifically if available. One concrete hook already present in the project’s own notes: DD-020 in `docs/decisions_es.md` reads the graph’s low density as potentially sociological rather than purely technical — a very sparse network as a possible sign of diaspora fragmentation in a host country with high individualisation, citing Vertovec (2009). That citation needs to be checked and properly sourced before use, and is currently the only literature reference recorded anywhere in the repo. **]**

2.2 Cultural-Sector and Creative-Industry Network Mapping

[TODO: New section (was folded into the diaspora section in the first outline). Needed because the project’s inclusion criterion is now cultural relevance, not national origin: cover prior work mapping cultural/creative-sector actors as networks (cultural intermediaries, venue/institution ecologies, scene and field theory, cultural participation and discoverability). This is the literature that justifies treating a heterogeneous set of venues, institutions, collectives and individuals as one object of study — and against which the project’s preliminary, since-superseded empirical observation on this point (a small institutional core amid a large organic periphery, §7.4.1) should be read, rather than treated as a settled finding. **]**

2.3 Social Network Analysis: A Brief, Deliberately Limited Treatment

[TODO: This section is short by design, not by omission — see §3.5 for why. It needs only enough to support two remaining live uses of network metrics: PageRank and betweenness centrality, defined precisely enough to justify their role as one term of the site’s ranking formula (§3.11) and one interface badge. A fuller grounding in SNA concepts (community detection and modularity, structural holes and brokerage, the participation coefficient, the E-I index) belongs with the closed exploratory analysis of §7.4.1 and should only be written at length if that line of work is actually resumed. **]**

2.4 Computational Social Science and Social Media as Data

[TODO: Justify Instagram as a data source; discuss known biases (who is and is not represented on Instagram, algorithmic visibility, platform terms of service, and the ethics of public-data research). Two project-specific points to connect to the literature: (i) the corpus mixes a behavioural signal (mentions, tags, co-authorship) with a platform-inferred one (Instagram’s own `relatedProfiles` suggestions); (ii) scraping is mediated by a commercial actor (Apify), which is itself a methodological dependency worth naming. **]**

2.5 Graph Databases as the Storage Model

[TODO: Neo4j and the property-graph model; why a graph database rather than a relational one for this problem (DD-001) — the argument is about the shape of the data (heterogeneous entities and relationships: accounts, posts, hashtags, locations, events) and about supporting the event-extraction and geocoding pipelines’ write patterns, not about running in-database graph algorithms, which §3.5 explains this project does not do.]

2.6 NLP for Event Extraction and Classification

[TODO: Ground the three-layer detection design: sentence embeddings for candidate filtering, zero-shot NLI classification, and LLM-based extraction, classification and gating. Cite relevant NLP literature (sentence embeddings and semantic similarity, zero-shot NLI, LLM information extraction, and the reliability problems of LLM output), and treat the multilingual requirement (ES/FR/EN) as a specific design constraint rather than an implementation detail — DD-008 records a monolingual English NLI model producing a 0.7% detection rate on Spanish and French captions, which is the empirical anchor for this argument.]

2.7 Positioning

[TODO: One or two paragraphs. Three differentiators are already supported by the material in this thesis: (i) against prior diaspora network studies — a data-driven, automated-plus-human-curated hybrid that produces a live public tool rather than only an analysis; (ii) against prior event-extraction NLP work — domain-specific gating logic validated by blind human comparison at three successive scales (§3.8), and the finding that the semantic reference set encoded a cultural bias measurable in recall; (iii) against both — the explicit replacement of a nationality-based inclusion criterion by a cultural-relevance one, including the removal of a nationality bonus from the ranking formula (§6.1).]

Chapter 3

Methodology

3.1 Research Design Overview

The project is organised as a seven-phase pipeline, each phase implemented as one or more independent, re-runnable scripts writing to a shared Neo4j graph:

1. **Extraction** — Instagram profiles and posts are scraped through Apify actors into local JSON (`1_harvest_ig_profiles.py`, `1_harvest_ig_posts.py`, `extract_profiles.py`).
2. **Ingestion** — the raw JSON is loaded into Neo4j Aura (`2_build_graph.py`).
3. **Network analysis** — the graph is exported to CSV and analysed offline with `igraph` and `leidenalg`, and the resulting metrics are written back (`3_analyze_network.py`).
4. **Manual categorisation** — a hand-curated spreadsheet of account attributes is uploaded onto the matching `:Account` nodes (`load_manual_account_categorization.py`).
5. **Event enrichment** — posts are passed through the three-layer detection pipeline to create `:Event` nodes, with inline deduplication (`4_enrich_events_extract.py`), followed by a standalone deduplication pass over existing events (`4_enrich_events_resolve.py`).
6. **Geographic enrichment** — location strings are geocoded and placed in an administrative hierarchy (`4_enrich_locations.py`).
7. **Cleanup** — accounts never touched by manual curation, and everything depending exclusively on them, are deleted (`cleanup_legacy_accounts.py`).

Two design principles run through all seven phases and are treated in this thesis as methodological rather than merely technical. *Idempotency*: every script can be re-run

without duplicating or corrupting what previous runs produced, because each checks a sentinel property before processing (`eventExtracted`, `lat`) or merges rather than inserts. *Separation of provenance*: the manual curation phase deliberately writes to a property namespace (`artType`, `culturalIdentity`, `geoZone`, `manualFollowersCount`) disjoint from the one the ingestion phase writes, so that neither can silently overwrite the other. The practical consequence is that automated and human-produced data remain distinguishable in the graph after the fact, which is what makes the curation-as-trust-signal argument of §3.9 verifiable rather than merely asserted.

[TODO: Insert the pipeline diagram here, adapted from `docs/data_flow.md` (an English version already exists alongside `docs/data_flow_es.md`). **]**

3.2 Research Operations: Decision Logging, Run Tracking, and Iterative Safety

Independently of the technical pipeline itself, this project maintained three interlocking practices for managing the research process, treated here as a methodological contribution rather than incidental project management. Together they make the pipeline’s evolution auditable after the fact — which is not free in a setting where the object of study is a continuously growing graph, reprocessed repeatedly by successive versions of the extraction logic.

A real-time decision log. Every non-trivial design choice was recorded, at the moment it was made, in a sequentially numbered decision log (`docs/decisions_es.md`, fifty-one entries numbered DD-001 to DD-053; two numbers in that range, DD-021 and DD-024, were reserved for decisions referenced in the code but never written up) rather than reconstructed retrospectively for this thesis.¹ Each entry states the alternatives considered, the evidence that motivated the final choice, and, where applicable, validation results. Four entries illustrate the range of what this captured. DD-041 documents a geographic-conflict guardrail added to the event-deduplication resolver after a production dry-run surfaced a specific failure mode: two structurally similar events merged despite being in different countries. It was validated first against fourteen isolated test cases and then in a live dry-run over the 664 events then in the graph, in which it blocked six geographically contradictory merges while leaving intact the 68 duplicate pairs the run had identified for merging. DD-042 documents the addition of a per-event, LLM-generated tag field, designed specifically to avoid a parsing failure present in an earlier, account-level free-text field, and scoped explicitly as prospective — the events already in the graph were left without it rather than backfilled at token cost. DD-043 and DD-044

¹The log is maintained in Spanish, the language in which the project work was conducted; `docs/decisions.md` holds a partial English translation that currently stops at DD-033. Where entries are quoted in this thesis, the translation is the author’s.

document the design of the Neo4j-to-static-JSON export and the resulting public website, including an explicit accounting of which ranking components could be precomputed at export time and which had to remain client-side because they depend on the current date or on the individual visitor’s session. Maintaining this log in real time, rather than writing it up after the fact, is itself a methodological choice: it captures the reasoning available *at the time* of each decision, including alternatives that were considered and rejected — something retrospective documentation systematically loses.

A run log. Separately, every substantive execution of the pipeline against live data — not every script invocation, but every run whose output was retained or acted upon — was logged with its date, parameters, and a summary of results (`docs/runs_log_es.md`, RUN-001 to RUN-021 at the time of writing). This served two purposes beyond record-keeping. It made it possible to distinguish a genuine change in the underlying data (new accounts or events discovered) from a change in pipeline logic when comparing two exports — a distinction that becomes impossible to reconstruct once several months of both have accumulated. And it created a paper trail for decisions that depended on a specific run’s output rather than on memory: the decision not to add a venue-specificity guardrail to the event resolver, for instance, is recorded in RUN-021 with the concrete case that prompted it (one event node absorbing three separate watch parties for the same football match, held in different Paris bars) and the reasoning for accepting it.

Dry-run-first, diagnose-first design. Every script that writes to the live Neo4j graph was built with a `-dry-run` or `-diagnose` mode as a first-class feature, not an afterthought: `cleanup_legacy_accounts.py` runs its deletion logic inside a single transaction and issues a `ROLLBACK` under `-dry-run`, so a preview’s reported counts are exact rather than estimated; `load_manual_account_categorization.py` and the event-extraction and event-resolution scripts all follow the same pattern. Combined with the idempotent design of the extraction and ingestion scripts, this made it possible to iterate on pipeline logic against a live, shared, non-trivially-sized research database without the usual risk of an irreversible mistake. The discipline was not self-enforcing, however, and DD-038 records what happens when it lapses: the diagnostic CSV export was gated on `-dry-run`, so the first real production run of the event extractor produced no comparison file and — because event timestamps were written as strings rather than native Neo4j datetimes — could not be reliably queried by date afterwards either. A run that worked correctly left almost no auditable trace. The fix restored parity between the two modes, on the principle that a production run needs to be at least as inspectable as a rehearsal.

Together, these three practices form a coherent research-operations discipline: decisions are recorded when made, runs are recorded when executed, and changes to the live data are always previewable before being committed. For a thesis built on a continuously evolving dataset rather than a fixed one, this discipline is what makes it possible to state, for any given result in Chapter 5, exactly which version of the pipeline and which run

produced it.

3.3 Data Collection

Data collection relies on Apify's hosted `instagram-profile-scraper` and `instagram-post-scraper` actors rather than on a self-maintained browser-automation stack (DD-002). The reasoning recorded at the time was one of maintenance cost and risk: Instagram actively blocks direct scraping, so a self-hosted scraper implies continuous maintenance and IP-blocking risk, whereas a hosted actor externalises both at a metered per-use price. That price is not incidental to the method — it is tracked per run in `.apify_cost_log.json` and functions as a real constraint on sampling decisions, which is why the harvesting logic is built around spending as little as possible per unit of new information. Profile scraping and post scraping are separate scripts (DD-003) because they use different actors, return different JSON structures, and need different refresh frequencies.

Seed selection changed across the project's two phases. The first phase used a single seed, the Colombian consulate in Paris, and expanded outward breadth-first through mentions, tags and Instagram's own `relatedProfiles` suggestions (DD-006); the reasoning was that a consulate is the most central institutional node of the community it serves, so relevant accounts should lie within one to three hops, and that organic expansion avoids the researcher-bias of a hand-written list. The second phase replaced the single seed with curated seed files (`config/seeds_*.json`) covering Latin American consulates and embassies in France plus French cultural and academic institutions connected to Latin America (DD-022). These institutional seeds raised a conflict the log resolves explicitly: the account-classification scheme treated government accounts as out of cultural scope, yet the entire discovery process depends on them. DD-026 separates the two questions — *is this account a valid target for cultural analysis?* and *is this account structurally necessary to discover the network?* — and assigns consulates the role of seed source without treating them as cultural targets.

Post collection is deliberately recency-biased rather than density-maximising. Rather than retrieving a fixed number of historical posts per account, the harvester computes a per-account time window equal to the gap since that account's most recent known post, capped at a configurable maximum, and merges results into a rolling window of the fifty most recent posts (DD-028, DD-029). The rationale is that the downstream objective is detecting forthcoming events, not reconstructing publication history; the earlier fixed-window design had a blind spot in which an account with a recent post was skipped entirely, losing anything published in between. A related robustness fix is worth recording as method rather than trivia: the Apify actor signals "no content found" by returning a one-element list containing an `error` key, which is `truthy` in Python and was therefore

silently treated as a successful single-post result, corrupting both the run counters and — before the merge logic existed — the stored file (DD-030). Data collection failures in this setting do not always announce themselves as failures.

3.4 Data Model

The graph uses eleven node labels and sixteen relationship types. Accounts, posts, hash-tags, locations, music tracks, comments and IGTV videos come from ingestion; events come from the NLP pipeline; cities, arrondissements and countries come from geocoding. Accounts additionally carry `:Public` or `:Private` as a second label rather than a boolean property (DD-005), which allows indexed filtering in Cypher and visual separation in the Neo4j browser. The full schema is reproduced in Appendix A.

The design principle behind the node set (DD-004) is that the model should capture three distinct dimensions at once: content (posts, hashtags), social interaction (mentions, tags, co-authorship), and cultural context (locations, music tracks). A minimal account-and-post model was considered and rejected on the grounds that it would discard exactly the signals the research question depends on — venues, collaborations and thematic markers. One consequence of this model matters for the analysis chapter and is easy to overlook: social relationships such as `MENTIONS` and `TAGS_USER` originate at the `:Post` node, not at the account, so any account-to-account social network has to be obtained by projecting `author→post→target` rather than read directly off the stored edges (§3.5).

3.5 Network Analysis: Status and Scope in This Thesis

An early phase of the project treated graph-theoretic analysis of the account network — exact PageRank and betweenness centrality, Leiden community detection, k-core decomposition, participation coefficient and E-I index, computed offline with `igraph` and `leidenalg` over two separate network layers because Neo4j’s Graph Data Science library was unavailable in this environment (DD-013) — as a central method and, through an automated classifier built on top of it, as the intended criterion for deciding which accounts belonged in the corpus at all. That role did not survive contact with the data: §3.9 reports that under 10% of the accounts the classifier selected turned out to be culturally valuable on manual review, and hand curation replaced it as the operative trust signal. Community detection, modularity and brokerage analysis are consequently no longer part of this project’s active methodology, and are not reported as findings in Chapter 5; the full design, together with the one substantive result it produced before being superseded — that some of the most structurally central accounts in the network had almost no content of their own, which is direct evidence for why a content-based selection criterion was the wrong one — is kept as a documented but closed line of work in §7.4.1.

One artefact of that earlier work remains live rather than archived: the account-level PageRank and betweenness values it computed are still read by the site’s ranking formula (§3.11, term *A*) and drive a single interface badge marking structurally bridging accounts. That residual dependency, and the caveat it carries — the values are not recomputed on an ongoing basis — is documented where it actually matters, alongside the formula itself, rather than here.

3.6 Event Detection Pipeline

Event detection runs in three layers over post captions. The design was originally four layers and was reduced to three in August 2026 after the evaluation described in §3.8.

Layer 1 — semantic candidate filtering. Each caption is embedded with `paraphrase-multilingual-MiniLM-L12-v2` and compared by cosine similarity against a reference set of roughly 100 phrases covering several event types, three languages and six announcement forms (calls, openings, listings, date-plus-place, registration, reminders). Embeddings rather than keywords were chosen because the corpus mixes Spanish, French and English with emoji and slang, where keyword matching fails on orthographic variation and indirect phrasing (DD-007); the reference set is large rather than minimal (DD-010) because a narrow reference space misses exactly the small community events the project most wants to capture. Layer 1 exists primarily as a cost control: running the NLI model over every post was measured at five to seven hours on the available hardware, whereas pre-filtering reduces the candidate set to roughly 40% of the corpus and the total to about eighteen minutes (DD-009).

Layer 2 — is this event-like at all. Surviving candidates are passed to a binary natural-language-inference classifier (`MoritzLaurer/multilingual-MiniLMv2-L6-mnli-xnli`). Two decisions here are worth stating explicitly. The model is multilingual by requirement, not by preference: an earlier English-trained NLI model produced a detection rate of 0.7% on Spanish and French captions — effectively noise (DD-008). And the hypothesis sentence is now formulated in the caption’s own detected language, after a fixed-Spanish hypothesis was shown to depress scores for English and French captions regardless of their content.

Layer 3 — LLM extraction, typing and gating. Candidates that pass Layer 2 are sent to a large language model — Ollama locally, or one of four cloud providers tried in order with automatic failover and a cap on rate-limit waiting (Groq, Google, Cerebras, DeepSeek, DD-033) — which extracts title, description, date, location and price range in Spanish, together with French translations of the title and description generated in the same call at no extra request cost (`title_fr`, `description_fr`, DD-051, feeding the bilingual site of §4.5), assigns an event type from a fixed sixteen-label taxonomy, proposes one to three free-form artistic tags (DD-042), and — the decisive change — *gates* node

creation. If the model reports that the post is not a public invitation, or not upcoming, no `:Event` node is created at all. In the earlier design the node was created regardless and merely scored lower, which §3.8 shows was the single largest source of false positives.

DeepSeek sits last in that order rather than third, and the reordering is itself evidence about a failure mode worth recording. In its original position it failed more often than the three free-tier providers, but not through the expected quota error: it returned HTTP 200 with an empty `message.content`, which the JSON parser downstream treated identically to a malformed response until an explicit emptiness check was added to tell the two failure modes apart (DD-050). Because DeepSeek is the only one of the four that is paid, and proved the least stable of the four across a live run of 315 posts, it was moved behind the three free providers so that it is tried only once they are all exhausted, rather than ahead of one of them for no reason the observed failure pattern could justify.

Three further constraints shape the layer-3 stage. Calls are deduplicated by post identifier within a run, so co-published posts share one call. Reasoning traces are requested only in dry-run mode, since they are expensive and only useful for auditing. And a deterministic `has_text_date` gate — requiring that a date actually appear in the caption text — is evaluated *before* the model is called rather than after, which was measured to remove 40.7% of calls offline and 48.3% in production without changing a single final decision (DD-036). The date extractor behind that gate is itself constrained to avoid manufacturing precision: a month or year mentioned without a day is discarded rather than resolved to an arbitrary first-of-month (DD-034), after four disagreements in the evaluation sample were traced to exactly that behaviour.

Newly created events inherit `artType`, `institutionType`, `culturalIdentity`, `geoZone` and `parentInstitution` directly from the publishing account’s manual curation, at no model cost, whenever that account has been curated. Deduplication happens twice: an inline resolver checks cosine similarity above 0.82 together with a location match and a date within three days before creating a new node rather than enriching an existing one, and a standalone resolver (§3.7) runs afterwards over the accumulated set.

3.7 Event Deduplication

Several accounts routinely publish about the same event, so deduplication is not an edge case but a structural feature of the corpus. The original resolver grouped events by normalised location string and compared pairs only within a group, using date proximity and cosine similarity as secondary criteria (DD-011). An audit of a production run showed this ordering to be the wrong way round: two posts about the same concert written the venue address differently and therefore never met, and an event with no location string at all could never be compared with anything (DD-040).

The redesign inverts the order. All pairs are compared by embedding similarity first,

using a vectorised cosine matrix, with dates as mandatory confirmation rather than as a primary filter: where both dates exist they must fall within three days and similarity must exceed 0.75; where one is missing, a stricter similarity threshold of 0.85 compensates for the lost corroboration. Location was explicitly *not* converted into an embedding, on the reasoning that text embeddings capture lexical rather than geographic similarity — two different theatres sharing the words "Théâtre de" may embed close together, while two addresses on the same street may not. An audit of the location strings then in the corpus supported demoting them: roughly a fifth carried a full address, over half only a venue name, a tenth only a city or country, and around a seventh were empty.

The predictable cost of dropping the location filter appeared on the first live dry-run: merges that were geographically contradictory, such as a Paris event merging with a Madrid one, because generic community posts share vocabulary regardless of where they take place. The response (DD-041) was a deliberately small gazetteer of country and city names drawn from the actual failure cases, which blocks a merge only when both sides name a recognisable and non-overlapping country. It is permissive by design — insufficient evidence on either side never blocks — precisely to avoid reintroducing the exact-match problem the redesign removed. Its residual limit is documented rather than patched: it compares countries, so two different venues within the same city can still merge, which RUN-021 shows happening and which the run log records as an accepted trade-off for the case at hand.

3.8 Evaluation Design: Blind Human-versus-Pipeline Comparison

The pipeline’s accuracy was assessed by blind manual classification of post captions against the pipeline’s own decisions, repeated at three increasing scales, with each round feeding concrete changes back into the code. In each round the captions were classified without access to the pipeline’s decision columns, and only then joined against the diagnostic CSV the extractor emits.

Round 1 — 100 posts (2026-08-09). Overall agreement was 58%. The asymmetry mattered more than the headline: 35 disagreements were posts the pipeline called events and the human classification did not, against 7 in the opposite direction. Of those 35 false positives, 29 already carried the layer-3 model’s own verdict that the post was neither a public invitation nor upcoming — the system had the information and did not act on it, because node creation was not conditioned on those flags, only scoring was. On this sample, the pipeline’s actual behaviour (45 true positives against 80 created nodes and 52 real events, recall 0.865, precision 0.5625) scores $F1 = 0.682$; simulating the gate raised precision from 56% to 90% and overall agreement from 58% to 86%. This single finding

is what turned gating from a scoring adjustment into a creation condition (§3.6).

The same round produced the bias finding referred to in §1.1. The reference phrases of layer 1 had been written entirely in the vocabulary of Colombian diaspora events. Splitting the sample by account theme, the false-negative rate was 4.3% for Colombian and Latin American accounts against 12.9% for general and French accounts — real events with explicit dates and venues, in French and English, discarded just below the similarity threshold. A criterion that was intended to be about cultural relevance was, at the level of implementation, partly about national vocabulary.

Round 2 — 495 posts (batches through 2026-08-12). Aggregate precision was 0.824 and recall 0.858 ($F1 = 0.841$). Three code changes followed (DD-036): eighteen additional reference phrases in an institutional and formal register, added after the three layer-1 false negatives in the sample were found to share a sober institutional tone absent from the reference space; the deterministic `has_text_date` gate, which was shown offline to block 15 of the 27 false positives in the full sample; and an explicit negative instruction in the layer-3 prompt excluding course registrations, enrolment and admission calls, which had been producing events with a structure indistinguishable from real ones. A fourth change was considered and rejected on the evidence: lowering the layer-1 threshold from 0.45 to 0.35 would have recovered seven real events at the cost of admitting 47 additional non-events, a signal-to-noise ratio of roughly 1:6.7, which is the wrong trade in a pipeline whose dominant error was already precision.

Round 3 — 150 posts in a live dry-run (2026-08-12, RUN-020). Agreement reached 96.7%, with precision 97.3% and recall approximately 90% ($F1 \approx 0.935$) — the first round run against the modified pipeline rather than a simulation. Four of the five remaining disagreements were traced to specific lines of code and fixed the same day (DD-037): a theatre season written "26/27" parsed as a day-and-month date, and a 600-character truncation window in the date extractor that discarded dates appearing later in the long multi-line captions typical of event flyers. The fifth had no identifiable code cause — two nearly identical posts about the same event, one accepted and one rejected by the language model — and is recorded as observed variability rather than resolved.

Two methodological caveats belong with these numbers. First, the human labels are the author's own, so this measures agreement with a single annotator's criterion, not with a consensus ground truth; the criterion itself had to be adjudicated in at least one case, where exhibitions announced with an explicit closing date were ruled to count as events even when the surrounding text is retrospective (DD-035), and that adjudication remains flagged as unresolved against the pipeline's own behaviour. Second, agreement rose across the three rounds partly because the pipeline improved and partly because the samples differ in composition, so the figures track a trajectory rather than three measurements of the same quantity.

3.9 Human Curation as the Trust Signal

The project's original design selected accounts automatically. A classifier scored each discovered account on geographic and cultural dimensions, combining semantic similarity against reference embeddings with hard signals from the profile, and accounts above a threshold were kept. In practice, well under 10% of the accounts it selected turned out to be culturally valuable on manual review. The response was not to tune the classifier further but to change what the pipeline treats as authoritative: the property `manualDataCuratedAt`, written only by `load_manual_account_categorization.py` when an account appears in the hand-curated spreadsheet, became the operative signal for "this account is worth keeping", and the automated score was retired from that role.

This is a pivot worth stating as a finding rather than as a confession. Three mechanisms visible in the project's own records explain why the automated criterion underperformed. First, the score rewarded signals that correlate with visibility rather than with cultural relevance, so commercial and institutional accounts with strong profile metadata scored well while small collectives with sparse profiles did not. Second, the geographic component was systematically wrong in a way that was invisible until profiles were inspected by hand: several accounts scored as France-based were physically located elsewhere, and the fix required reading latitude and longitude out of the business address and testing them against a bounding box (DD-031), followed by a further narrow rule for a residual class the bounding box could not catch — Alliance Française branches in Latin American cities, whose bios legitimately mention France as a subject rather than a location (DD-032). Third, and most tellingly, an account's structural importance in the network is not the same quantity as its content volume: the one-time network analysis of §7.4.1 found accounts ranking among the top bridging and top PageRank actors while having almost no posts loaded, their centrality coming entirely from being mentioned by others. Any selection criterion that reads relevance off an account's own output will miss these.

Curation also had to be reconciled with discovery, since the ingestion step creates one account node per Instagram `relatedProfiles` suggestion on every scrape. Treating "not curated" as "delete" would have destroyed fresh discoveries alongside old noise. The cleanup script therefore tags, rather than deletes, any uncurated account reachable via `RELATED_TO` from a curated one, marking it `discoveredViaCuratedAccount` with a pending review status, so that these accounts survive as a review queue. The cost of the approach is stated plainly in §3.14: human curation does not scale, and the boundary of the corpus is therefore set by how much reviewing one researcher can do.

3.10 Geospatial Enrichment

Location strings extracted by the event pipeline are geocoded with Nominatim, the Open-StreetMap geocoder, chosen because it is free, requires no API key, covers both France and Colombia adequately, and imposes a rate limit (one request per second, implemented here as 1.1 seconds between requests) that is tolerable at this corpus size (DD-018). Resolved locations are placed in an administrative hierarchy — **Location** → **Arrondissement** → **City** → **Country** — with Paris arrondissements derived from postal codes in the 750XX range. Sessions are logged so that a partial run can be resumed rather than repeated.

The methodologically interesting part is what happens when geocoding fails. Roughly one event in seven never acquires coordinates, for two distinct reasons that should not be conflated: some have no location text in the caption at all, and some have text that Nominatim cannot resolve. A fallback to a city-level hint was available and was rejected as a display strategy, because it produces coordinates that look precise and are wrong — an event placed at the centre of Paris because its address could not be parsed is indistinguishable, on a map, from an event that really is there. The decision taken (§4.5) is that the discovery site shows only events with genuinely resolved coordinates. This is a presentation filter, not a deletion: the **:Event** node remains in the graph for analysis, it simply does not enter a product whose central organising principle is geography.

3.11 Ranking and Relevance Scoring

The discovery website ranks events with an explicit, documented formula rather than an opaque model, on the stated principle of contextual algorithmic curation rather than personalisation: there are no user accounts, and the same conditions produce the same ordering for every visitor. All component signals are converted to percentile ranks over the universe of publishable events before being combined. The score is

$$\text{Relevance} = 100 \cdot (0.30 Q + 0.22 A + 0.18 B + 0.20 T + 0.10 C) \cdot P$$

where Q is detection quality ($0.70 \cdot \text{eventScore} + 0.30 \cdot \text{confidence}$); A is the authority of the publishing account in the network, itself a weighted blend of PageRank (0.40), betweenness (0.25), follower count (0.20), participation coefficient (0.10) and verified status (0.05); B is social resonance (hotness score and post count); T is temporal proximity, implemented as explicit steps rather than an exponential decay so that it can be read and defended; and C is session context. P is a multiplicative penalty or bonus: political accounts $\times 0.55$, low-confidence extractions $\times 0.80$, free events $\times 1.05$.

Three aspects of this formula are methodological choices rather than tuning. First, follower count is deliberately the smallest term in A : a consulate with forty thousand followers is not thereby more culturally relevant than a collective with three thousand,

and PageRank is weighted highest precisely because it measures recognition *within the cultural network* rather than general reach. Second, Q , A , B and P are precomputed at export time because they require percentile context over the whole dataset, whereas T and C are computed in the browser on each visit because they depend on the current date and on the individual session — precomputing them would freeze the ranking to the last export (DD-043, DD-044). Third, and discussed at length in §6.1, an earlier version of the formula included a $\times 1.05$ bonus for any event carrying a non-null `culturalIdentity` value; it was removed when the project’s scope broadened. A mandatory diversification pass then caps how many events from the same author, category or parent institution can appear together at the top of the home page, without which the largest category mechanically dominates the front page.

A caveat belongs here rather than only in §3.5: the PageRank and betweenness values behind A are not recomputed on a running basis. They are read off whichever account happened to be covered by the last executed pass of the offline network analysis, so an account curated afterward, or a shift in who mentions whom since that pass, is not reflected in A until the analysis is deliberately re-run. This is a live product dependency on a component this thesis otherwise treats as a closed, superseded line of work (§3.9); it is reported honestly here rather than smoothed over, precisely because it is the one place where that earlier work still has a measurable effect on what a visitor sees.

3.12 Publication Lifecycle and Event Retirement

Detection and ranking are only two stages of a longer cycle that this thesis treats as methodology rather than as an implementation footnote, because it is what makes the published catalogue trustworthy over time rather than only accurate at the moment of extraction. Three mechanisms close that loop, in the order an event actually passes through them.

Staged human review. Every `:Event` node created by the extraction pipeline (§3.6) is tagged `:PendingReview` at the moment it is created, and both the JSON export that feeds the website and the Excel export used internally exclude that label by default. Concretely, this means the layer-3 gate (§3.6) is a necessary but not sufficient condition for publication: passing it only qualifies an event to enter a review queue, not to appear on the site. A small local application (`review_events.py`) lets the researcher approve, edit or reject each pending event; rejection is deliberately soft — it adds a `:Rejected` label rather than deleting the node — so that no review decision is irreversible (§4.5, DD-047). This reuses the same label-based status pattern already used for the automated gate rather than introducing a second status system for human decisions, and it is the direct methodological answer to §5.3’s finding that twelve of twenty highest-resonance institutional events were, on manual inspection, either not events at all or misclassified:

automated gating alone was not sufficient to keep the published catalogue clean, and a second, human-executed gate was added downstream of it rather than tuning the first one further.

A rolling retention window, not deletion. Consistent with the project’s broader practice of never deleting research data outright (rejection above, and the researcher-only tier of the Excel export), `:Event` nodes are never removed from the graph once approved. Instead, retention is enforced only at export time: the export query excludes any event more than a fixed number of days in the past (thirty by default, configurable per run), and the website’s own client-side logic enforces the same cutoff a second time as a deliberate redundancy, so that a stale export cannot silently resurrect old events even if the cutoff at generation time were ever bypassed. The researcher-facing Excel export is exempt by design: it remains the full historical archive for analysis, while the public site only ever shows what is current. Before this cutoff existed, 71% of parseable events in the graph were already more than thirty days old and every one of them was still being shipped to every visitor; the fix is a filter on read, not a change to what is stored, which keeps the analytical record in §5 intact regardless of what the live site happens to be showing on a given day.

Publishing is a deliberate, separate step from generating data. Regenerating the exported JSON file, regenerating the static search-engine-facing pages built from it (§4.5), and actually deploying the site to its hosting platform are three distinct manual commands, none of which triggers another. This was not the original design intent so much as a gap discovered in production: an approved batch of events did not appear on the live site because the data had been re-exported but never redeployed, and the site did not look broken, only stale, which is precisely why the gap went unnoticed for a period (DD-053). The lesson recorded alongside the fix is general rather than specific to this project: a pipeline stage that succeeds silently and a pipeline stage that was never run produce an indistinguishable symptom — nothing changes — unless the two are made observably different, and collapsing the three steps into one command is listed as future work (§7.4) for exactly that reason.

3.13 Ethical Considerations

The corpus is restricted to public accounts and publicly visible posts; private accounts are labelled as such in the graph and are not scraped. This does not by itself settle the ethics, since public availability and consent to research use are different things. Three positions were taken in consequence. Organisational accounts — venues, institutions, collectives — are treated as public actors and named directly, since their posts are promotional by intent and naming them is what makes the mapping useful to the community. Individuals appearing incidentally in captions, comments or tags are not the object of analysis,

and the project’s data-retention policy works in their favour: the cleanup phase deletes accounts never touched by manual curation together with their posts and comments, so the accumulated periphery of individual accounts discovered through mentions is removed rather than retained indefinitely.

The public artifact adds a second set of considerations, handled by design rather than by policy. The site stores no personal data on any server: it is fully static, session preferences live in the browser’s `localStorage` under a single key with no third-party cookies and no personally identifying information, and a reset control is exposed in the interface. Event cards link back to the original Instagram post rather than reproducing it, and the site presents itself as a discovery tool rather than a ticketing service. Automated detection is disclosed on each event, which matters because a language model sits in the pipeline and, as §3.14 documents, is wrong a measurable share of the time.

[TODO: Two items still to settle before submission: (i) whether the university requires a formal ethics statement or data-management plan for public-web data, and in what form; (ii) the precise status of Instagram’s terms of service with respect to scraping via a third-party actor, stated accurately rather than assumed — this is a legal-risk question the thesis should address explicitly rather than leave implied. **]**

3.14 Limitations of Method

The limitations below are drawn from failures actually observed in this project’s runs, not from a generic list of caveats.

Geocoding produces false precision when it fails. Around one event in seven has no usable coordinates, and the available fallback — resolving to a city-level hint — yields plausible-looking but wrong positions. The mitigation is exclusion from the map rather than approximation (§3.10), which trades coverage for honesty.

Date extraction is the pipeline’s most fragile component. Successive rounds of evaluation found dates invented from a bare month (DD-034), a theatre season "26/27" parsed as a calendar date, dates lost because the extractor truncated captions at 600 characters (DD-037), and implausible dates surviving a clamp that was silently skipped whenever either of two date parses threw — producing events dated 2036, 2052 and 2090 from text that mentioned an anniversary, an age limit and a duration in minutes (DD-039). Each was fixed once found; the pattern worth reporting is that all of them failed silently, and all were found by manual inspection rather than by any automated check.

LLM classification is inconsistent at the margin. RUN-020 recorded two nearly identical posts about the same event receiving opposite verdicts, with no identifiable code cause. Layer 3 is also capable of typing an event with a label meaning "not a cultural event" while the surrounding logic created the node anyway, because creation did not consult the category field (DD-039).

Instagram-only sourcing bounds what can be observed. Organisations that announce primarily through mailing lists, their own websites or other platforms are absent by construction, and the platform’s own visibility dynamics shape which accounts are discoverable through `relatedProfiles` at all. The algorithmic layer of the network is a proprietary artefact and is analysed separately for that reason (§3.5).

Curation does not scale, and the tier labels are sparse. The trust signal that replaced automated scoring is one researcher’s judgement. The consequence is visible in the network analysis: at the time of RUN-015 only 62 of 7,269 account nodes carried an explicit tier label, so tier-based comparisons describe a small hand-curated subset rather than the graph as a whole, and this must be stated wherever such comparisons appear.

Geographic scope leaks through automatic discovery. Accounts outside the intended geography enter the graph through `relatedProfiles` before curation catches them — Alliance Française branches in Colombian cities being the recurring case (DD-032) — and at least one event in the institutional sample examined in §5.3 was a restaurant menu in Spain with no connection to the project at all.

A property inherited from the account is used as a proxy for the event’s own location. Every `:Event` node inherits `geoZone` from the account that published it, at no additional extraction cost, and that inheritance has since become an operational filter: the human-review interface groups pending events by it for bulk triage (§4.5, DD-052). A Paris-based account can announce something happening elsewhere, however, so the property describes who posted, not necessarily where the event is. That the first bulk-filtered batch matched the reviewer’s manual judgement in practice is evidence the heuristic works for the common case, not evidence that it is correct for every event it groups.

Chapter 4

System Implementation: The Hub Cultural Platform

4.1 Pipeline Architecture Overview

The system is a set of independent command-line scripts sharing a Neo4j Aura database, rather than an orchestrated application. Each script owns one phase (§3.1), reads what it needs from the graph or from disk, and writes back a disjoint set of properties. There is no scheduler and no continuous integration: runs are manual and deliberate, which suits a research setting where each execution costs money (Apify credits, LLM tokens) and where the point of a run is usually to test a change rather than to keep a service fresh.

Two properties of this architecture are load-bearing. *Incremental processing:* `extract_profiles.py` queries the graph for accounts not yet scraped rather than working from a static list, and the NLP phases skip any node whose sentinel property is already set, so a re-run costs nothing for work already done and an interrupted run can simply be restarted — which is how the connection failure mid-way through RUN-021 was recovered. *Namespace separation:* manual curation and automated ingestion write property names chosen not to collide, so re-running either never clobbers the other’s data.

4.2 Extraction and Ingestion

Extraction writes one JSON file per profile into a gitignored `data_raw/` directory; ingestion reads that directory and merges its contents into the graph. Ingestion is written entirely with `MERGE` rather than `CREATE`, and stamps every node it touches with `firstSeenAt` on creation and `lastUpdatedAt` on every write, which gives a per-node provenance trail without a separate audit table.

An instructive dead end is worth recording. Nodes created before that provenance stamping existed have neither property, and a script (`seal_legacy_batch.py`) was writ-

ten to tag them as a legacy batch so they could be identified later. By the time it ran, every node already carried some other date field, so it had nothing to seal and the `legacyBatch` property never marked a single real node. The mechanism that actually solved the underlying problem — separating material worth keeping from accumulated noise — was not a timestamp at all but the curation signal (§3.9). The lesson generalises: provenance metadata answers "when did this arrive", which turned out not to be the question that needed answering.

4.3 Network Analysis Engine

`3_analyze_network.py` is the component behind the closed line of work described in §3.5: it exports nodes and edges to CSV, builds two network layers in `igraph`, computes exact PageRank, exact betweenness and Leiden community detection, and writes results back with batched `UNWIND` statements, entirely offline from the exported CSVs with no live database connection required. It is documented here, briefly, only because two of its outputs — PageRank and betweenness per account — are still consumed downstream by the export script (§4.5) and therefore still reach the live site (§3.11); it is not run on any regular cadence, and the account coverage it provides reflects whichever pass last touched a given account rather than the current graph.

4.4 Event Enrichment Services

The event phase is where the system depends on components it does not control: two sentence-transformer models, an NLI model, and a hosted language model. Robustness therefore had to be built at the boundary. The extractor selects between a local model server and two cloud providers, marks a provider as exhausted for the remainder of a run once it has failed, and switches to the alternative rather than retrying indefinitely; a cap on rate-limit waiting prevents a provider's `Retry-After` header from stalling a run for minutes at a time (DD-033). The mechanism was validated by two distinct failure modes in practice — quota exhaustion and genuine network errors — and handled both by switching providers within ninety seconds.

4.5 The Discovery Website

The public artifact is a static site: `5_export_dashboard_data.py` reads the graph and writes a single JSON file, and `site/` contains hand-written HTML, CSS and vanilla JavaScript that consumes it, deployed to Cloudflare Workers as a static-assets bundle (`site/wrangler.jsonc`, Workers Assets). There is no backend, no framework and no

build step, which is a deliberate consequence of the dataset's size — the publishable events fit comfortably in a file of a few hundred kilobytes — and of the deployment constraint that the site must be reconnectable from a plain repository without configuration (DD-043, DD-044). The export is read-only and can be re-run after any extraction or resolution pass to refresh the data file, but regenerating that file and publishing it are two separate manual steps: neither a `git push` nor running the export script triggers `wrangler deploy`, a gap that went unnoticed for some time because the site had already been deployed once with older data and so never looked broken, only stale (DD-053).

Six design decisions are worth reporting as part of the thesis's applied contribution.

The category menu is fully dynamic. There is no editorial list of families hard-coded anywhere. The "what" axis is assembled on each load from the union of the event categories and per-event artistic tags actually present in the data, with real counts, ordered by volume. The menu therefore reflects what the upstream human curation has prioritised rather than an assertion in the code, and it changes shape as the curation does.

Recommendation without accounts. Session context is stored in `localStorage`: category weights increment when a visitor filters by or opens an event in that category, decay daily, and are dropped below a floor; recently seen events are down-weighted so the home page rotates. This yields the C component of the ranking formula with no login, no server and no personal data.

Similarity without exposing embeddings. The export computes cosine similarity across all events and stores, per event, the identifiers of its five nearest neighbours. The 384-dimensional vectors themselves are discarded after use and never reach the client, which keeps the payload small and avoids publishing a representation of the corpus that was never intended to be public.

Network analysis stays under the hood. PageRank, betweenness and community structure influence the ordering of events and occasionally surface as a badge, but the first version deliberately has no dedicated organiser-profile or methodology pages. This was a scoping decision: those screens are more legible as thesis material than as product, and the remaining screens (Explore, Map, organiser profiles) are listed as future work in §7.4.

Publication is gated by human review, not by the extraction pipeline alone. Every `:Event` node is created with a second label, `:PendingReview`, and both the JSON export and the Excel export exclude it by default, so a newly detected event stays invisible on the live site until it has been triaged. Triage happens in `review_events.py`, a small local Streamlit application with three actions per event — approve (removes the label), edit (an inline form that rewrites fields directly on the node), and reject — where reject is deliberately soft, adding a `:Rejected` label rather than deleting the node, so a mistaken rejection is always recoverable (DD-047). This reuses the same label-based status pattern already used for the automated Layer-3 gate (§3.6) rather than introducing a second status system for human decisions. In practice, the first real batch of pending events showed

that most out-of-scope items shared one property — a `geoZone` outside France, inherited from the publishing account at creation time — and reviewing those one by one was disproportionate to the judgement each one required, so the interface was extended with a filter on that property and a bulk-reject action scoped to whatever the filter currently shows (DD-052). The trade-off is explicit and unresolved: `geoZone` describes the account, not the event, so the bulk action is a heuristic for the common case rather than a per-event guarantee, and it remains at the reviewer’s discretion to apply.

Event content is bilingual only going forward, by design. The interface text — menus, filters, labels — already switched between Spanish and French; the content of individual events did not, because `title` and `description` were written once, in Spanish only, by Layer 3 (§3.6). Rather than translate the existing corpus after the fact, French versions are now generated by the same LLM call that extracts the event (DD-051), and the frontend falls back to the Spanish text whenever the French field is absent rather than leaving a blank card. The scope was deliberately narrowed to new events only, so the two language modes are not at parity and will not be until either enough of the corpus is re-extracted or a backfill pass is run; translation quality has also not yet been separately validated.

[TODO: Include two or three screenshots here rather than only in Appendix D, once the site has been reviewed in a real browser — DD-044 records that only the pure logic of `site/app.js` has been tested in isolation, not the rendered result. **]**

Chapter 5

Results

This chapter reports results from two sources: the event extraction and resolution of RUN-021 and subsequent runs, cross-checked where possible against the live export (`site/data.json`, 751 `:Event` nodes at the time of writing, prior to the retention filter of §3.12); and the evaluation trajectory of §3.8. Because the graph grows continuously, every figure below is stated with its date or run of origin rather than as a standing fact. A one-time structural snapshot of the account network — node and edge counts, community detection, bridging actors — was also produced (RUN-015, 2026-07-24) during the phase of the project described in §3.5, before the scope broadened and before the curation-based cleanup ran; it is not reported here as a current result, since neither precondition still holds, and is instead kept as a documented but closed finding in §7.4.1.

5.1 Thematic and Event Landscape

At RUN-021 the graph held 664 `:Event` nodes, of which 559 met the validity criteria for publication and 170 passed the additional requirement of resolved coordinates and entered the 2026-08-14 dashboard export alongside 200 accounts.

The distribution of the exported events by category is markedly uneven: gastronomy 51, training and workshops 27, institutional 22, visual arts 19, music 15, performing arts 9, community 9, festivals 6, audiovisual 6, politics 3 and academic 3. Gastronomy is thus the single largest category, at 30% of the exported set and 35% of the full graph.

This is a finding about the pipeline, not about the cultural field. The manual curation actively avoids gastronomy in favour of the project's declared priorities; the curator marks such accounts explicitly as "not applicable" rather than assigning them an artistic type. Gastronomy dominates because gastronomic posts are easy to scrape and easy to detect as events — they announce dates, venues and prices in exactly the register the detection layers reward — not because the project set out to collect them. The response was to demote the category in the interface rather than to filter it out of the data, keeping it available as a filter while removing it from the default prominence it would otherwise take

purely on volume.

Against the four declared thematic pillars, the picture is mixed. Table 5.1 shows the distribution of curated accounts and their events by artistic theme.

Table 5.1: Curated accounts and their events by artistic theme (2026-08-13, derived from the curated `artType` field).

Theme	Accounts	Events
Visual arts	21	25
Music	23	24
Dance	21	20
Theatre	15	16
Literature	15	7
Circus	5	5
Photography	3	2
Cinema	11	2

Two of the four declared priorities — photography and cinema — have curated accounts but almost no detected events, while visual arts, music and dance, which were never declared priorities, dominate by volume. The gap between eleven curated cinema accounts and two detected cinema events is a detection problem rather than an absence of cultural activity. A related structural gap was found in the taxonomy itself: none of the sixteen event types available to the extractor corresponds to literature, so a book launch or a poetry reading has no correct label and falls into an adjacent category or none. Since literature is one of the project’s four founding pillars, this is a pipeline gap rather than an interface detail, and it motivated the addition of per-event artistic tags (DD-042) — which, being prospective only, do not yet cover the events already in the graph.

The catalogue has grown substantially since that snapshot. As of the current export (`site/data.json`, prior to the retention filter of §3.12), the graph holds 751 `:Event` nodes across 349 curated accounts. The category-level picture is consistent with the RUN-021 finding rather than a correction of it: gastronomy remains the single largest category at 210 events (28.0%), followed by institutional (132, 17.6%), visual arts (87, 11.6%), training and workshops (74, 9.9%), community events (57, 7.6%), music and audiovisual (44 each, 5.9%), performing arts (34, 4.5%), academic (29, 3.9%), festivals (27, 3.6%) and politics (13, 1.7%). Only 5.6% of events in this snapshot are marked free, which is a property of what gets announced with a price attached in a caption, not necessarily of the underlying cultural field. Community events — the fifth, less formally bounded priority of §1.3 — have grown to 57 events, a large enough share (7.6%) to be worth tracking as its own line rather than folding it silently into the count.

[TODO: Re-run the theme distribution after a full extraction pass with `eventArtTags` populated, and report whether the photography/cinema gap narrows against the current, larger catalogue.]

5.2 Geographic Distribution

Of the 559 events meeting the validity criteria on 2026-08-13, 528 (94.5%) had an associated `:Location` node and 478 (85.5%) had real coordinates. The two shortfalls have different causes: 31 events carry no location text at all, because the caption never stated one, and 50 have location text that either failed to resolve in Nominatim or had not yet been through a geocoding pass. Only the first group is irrecoverable without a change to the extraction prompt.

Against the larger, current catalogue of 751 events, the shortfall is wider rather than narrower: 91.7% carry some location text (a venue name or address, short of a full coordinate pair), but only 47.4% have resolved coordinates. The gap between those two figures — text present but unresolved — is the geocoding backlog rather than a caption-level absence, consistent with an intermittent Nominatim failure observed on a subsequent full run (zero coordinates resolved out of 276 attempted, cause unconfirmed) rather than with a change in how often captions state a location at all. Because the discovery site excludes ungeocoded events from anything geography-dependent rather than approximating their position (§3.10), this backlog directly determines how much of the catalogue is usable in a geography-first product, and geocoding coverage is consequently one of the more consequential, and cheaper to fix, current data-quality gaps.

[TODO: Diagnose the zero-coordinate Nominatim run rather than only counting its effect — the geocoder script already sets a compliant custom `user_agent`, which rules out the most common cause of a blanket block, so the actual HTTP response has not yet been inspected. Once fixed, restate this coverage figure and report the distribution across arrondissements and communes, which the geocoding hierarchy supports but which has not yet been queried — this is the natural place to substantiate the Île-de-France scope claim of §1.3 with actual numbers. **]**

5.3 Data Quality Findings

Data quality is reported here as a result rather than deferred to a limitations footnote, because the error analysis is one of the more transferable outputs of the project.

The evaluation trajectory described in §3.8 is the headline: agreement between the pipeline and blind human classification moved from 58% (100 posts, 2026-08-09) through 82.4% precision and 85.8% recall (495 posts, 2026-08-12, $F1 = 0.841$) to 96.7% agreement with 97.3% precision and approximately 90% recall (150 posts, RUN-020, $F1 \approx 0.935$). The largest single gain came from one conditional statement — making node creation depend on the language model’s own invitation and upcoming flags, which it had been computing and ignoring — and it alone accounts for most of the roughly nine-point F1 improvement between the two rounds; the remaining, smaller gain came from the

reference-set and prompt changes made alongside it (§3.8).

A manual review of a targeted sample sharpens the picture where the aggregate figures are most flattering. The twenty institutional events with the highest resonance scores were classified by hand: roughly eight were genuine public events correctly categorised; roughly seven were not events at all — interviews, podcasts, press releases, appointment announcements, a private meeting — and had passed the public-invitation gate that should have stopped them; and roughly five were real events placed in the wrong category, including two visual-art installations, a concert tour, a restaurant menu in Spain unrelated to the project, and a language-course enrolment that an explicit prompt instruction was supposed to exclude. Twelve of twenty problematic is a much worse rate than the aggregate figures suggest, and the discrepancy is itself the finding: the sample was selected by highest resonance within the institutional category, which is precisely where the pipeline’s two known weaknesses concentrate — a leaky invitation gate and a tendency to type an event by the nature of the account that published it rather than by the content of the post. Aggregate accuracy over a random sample and accuracy within a targeted high-visibility slice are different quantities, and a discovery product surfaces the second.

Deduplication performance is measurable on the same run. Over 664 events, the resolver evaluated 220,116 pairs in roughly twenty seconds, identified 68 duplicate pairs, and consolidated them through 189 relationship redirections, transitive chains included. Of the pairs it rejected, 12,375 failed on similarity, 165,018 on the date window, and 6 on the geographic-conflict guardrail — the last group being precisely the failure cases that motivated the guardrail (§3.7).

[TODO: Two quantifications are still missing and would strengthen this section: (i) how many events had false-precision coordinates before the geocoding-confidence decision, obtainable by comparing a geocoding run with and without the city-hint fallback; (ii) an error rate for date extraction on the current code, since all the date figures cited above predate the DD-037 and DD-039 fixes. **]**

Chapter 6

Discussion

6.1 From a Diaspora Network to a Cultural-Relevance Network

The project’s scope changed twice. It began as a network of Colombian diaspora accounts, seeded from a single institutional account, the Colombian consulate in Paris. As the network grew through automatic relationship discovery (`RELATED_TO`), it broadened to Latin American diaspora accounts more generally. Both of these were still nationality-based criteria: an account or event qualified because of *who* was behind it.

The second, more consequential change replaced that criterion with a purely cultural one: an account or event qualifies if it is culturally relevant, with explicit priority given to photography, cinema, literature, theatre, and locally-organized community events, regardless of the nationality or cultural origin of the organizers behind it. This is a stricter methodological change than it might first appear, because it also required removing a bonus that had previously been present in the discovery website’s relevance-ranking formula (§3.11) — a multiplicative $\times 1.05$ applied to any event carrying a non-null `culturalIdentity` value, a field which in this corpus is overwhelmingly populated with Colombian and Latin American values. Retaining that bonus after the scope change would have produced a ranking signal that was formally neutral (it ranks *cultural* events) while materially still favouring one set of national identities over every other one present in the same graph; removing it was necessary for the ranking formula to be honest about what it actually measures. The field itself was kept, as an optional filter a visitor can choose to apply — the distinction being between an identity that a user selects and one that the system applies on their behalf without telling them.

The same audit produced a negative result worth recording: community detection was confirmed to be purely topological, computed only from graph edges, so `culturalIdentity` never entered the community structure through a back door. The bias in the ranking formula was explicit and removable; the more interesting bias was not.

That one is the reference set of the event detector’s first layer. Its roughly 100 phrases had been written entirely in the vocabulary of Colombian diaspora events, and the blind evaluation of 100 posts (§3.8) measured the consequence: general and French accounts lost real events at 12.9%, against 4.3% for Colombian and Latin American accounts — nearly three times the rate, on events with explicit dates and venues that were unambiguous to a human reader. A project that had already decided, at the level of stated policy, to include cultural activity regardless of origin was still, at the level of implementation, seeing that activity less well. This is the strongest argument in the thesis for treating the scope change as methodological rather than administrative: the nationality criterion was not only a boundary drawn around the corpus, it was embedded in the instrument that produced the corpus, where it was invisible until measured.

[TODO: Once network composition is re-pulled for Chapter 5, quantify here: what share of curated accounts and events carry Colombian, other Latin American, French, or other `culturalIdentity` values, before versus after the scope change. The account-level query is straightforward against the curated fields; the before/after comparison requires the archived run directories under `data_processed/runs/`. **]**

What this change gained, beyond ranking-formula honesty, is a criterion that is easier to defend and to operationalize consistently: *is this cultural, and does it fall within one of these priority domains* is a judgment that can be applied uniformly to any account the discovery process surfaces. *Is this part of the Latin American diaspora*, by contrast, required an ad hoc call for every borderline case — a French venue that regularly hosts Colombian artists, a Latin American cultural institute whose visitors are mostly French — that the pipeline had no principled way to resolve consistently. What it cost is some of the project’s original sociological specificity: the initial research question was, in part, about how *one* diaspora organizes culturally in a host country, and that question is necessarily diluted once the object of study is cultural activity in general. This thesis handles that trade-off by keeping the diaspora-specific reading as one lens applied to a subset of the network (wherever `culturalIdentity` data supports it), rather than as the network’s defining boundary.

6.2 On Data Quality and the Limits of Automation

The project’s most consistent lesson is that automated signals fail silently and confidently. This is not a general complaint about automation but a specific, repeated pattern in these results, visible in four independent places. The account classifier assigned high scores to accounts in the wrong country, because the geographic signal it trusted did not contain the field it assumed (DD-031). The geocoder, offered a fallback, returned coordinates that were precise and wrong rather than absent (§3.10). The date extractor filled in a day of the month that no caption ever stated (DD-034), and later accepted a theatre season as

a date (DD-037). The language model typed events with a label meaning "not a cultural event" while the surrounding code created them anyway (DD-039). In none of these cases did the system report low confidence; in every case the failure was found by a human reading the output.

The corresponding lesson about human curation is more qualified than "humans are better". Human curation was the more reliable trust signal here, and replacing automated account scoring with it was the right decision on the evidence — under 10% of automatically selected accounts survived manual review. But it does not scale, it embeds a single researcher's judgement, and the evaluation protocol that validated the event pipeline measured agreement with that same judgement rather than with any external ground truth (§3.8). What the project actually demonstrates is narrower and more useful: automation was most reliable where it was constrained to deterministic, checkable operations (does this caption contain a date; do these two coordinates fall inside this bounding box; do these two country names conflict) and least reliable where it was asked to make the judgement of relevance itself. The design that emerged puts deterministic gates in front of and behind the probabilistic components, and reserves the relevance judgement for a person.

[TODO: Extend with the implication for comparable projects, stated as design guidance rather than as a moral: put a cheap deterministic gate in front of any expensive probabilistic component, prefer absence to plausible-looking approximation in any user-facing field, and instrument production runs at least as well as rehearsals (DD-038). **]**

6.3 Practical Implications: The Website as Intervention

The discovery site is not only a dissemination channel but, in a modest sense, a sociological intervention: it makes a set of events visible together that were previously legible only to whoever already followed the right accounts, and its ranking formula makes an explicit editorial claim about what counts as relevant rather than leaving that claim implicit in an opaque feed algorithm. Two design decisions from the implementation are worth reading this way rather than only as engineering choices. The category menu is assembled dynamically from whatever themes and tags are actually present in the curated data, with no editorial list hard-coded anywhere (§4.5); the taxonomy the visitor sees is therefore a direct, live reflection of what the curation process has prioritised, not a claim fixed in advance and left to drift out of sync with it. And the removal of the nationality-linked ranking bonus (§6.1) is a worked example of an ordinary product decision — a multiplier in a formula — that was also, on inspection, a political one: a signal that looked neutral on its face was quietly favouring one set of identities over every other one present in the

same graph, and fixing that required treating a scoring weight as a claim about fairness, not only about relevance. Whether the artifact met the project’s fourth, deliberately human objective — leading to actually meeting someone through it (§1.3) — is addressed on its own terms in §7.5, since it is not a question a usage metric can answer.

6.4 Methodological Reflections

Four things would be done differently starting over, each supported by a specific failure already documented in this thesis rather than by hindsight in general. First, the human-curation trust signal (§3.9) would be built in from the start instead of arrived at after an automated classifier had already run long enough to accumulate a network’s worth of data around a criterion later judged wrong under 90% of the time. Second, the semantic reference set behind event detection would be written to the scope the project intends to end up with, not the scope it starts from — §6.1 shows that an inclusion criterion believed to be conceptual was in fact encoded, measurably, in a list of a hundred phrases, where it went unnoticed until a blind evaluation surfaced it. Third, production runs would be instrumented for audit from the first run, not after one proved unauditable (DD-038): the discipline described in §3.2 was a response to a specific incident, not a plan followed from day one. Fourth, every field a person will eventually read would default to explicit absence rather than a plausible-looking fallback — the geocoding, date-extraction and classification failures of §5.3 share this one shape, a system reporting something rather than nothing even when it does not know, and each was found only by a human noticing the output was wrong, never by the system flagging its own uncertainty.

Chapter 7

Conclusion

7.1 Summary of Contributions

The methodological contribution is the pivot from automated to human curation as the account-level trust signal (§3.9), evidenced rather than asserted — under 10% of automatically-selected accounts survived manual review — together with the blind human-versus-pipeline evaluation protocol that drove three successive redesigns of the event-detection logic and moved measured agreement from 58% to 96.7% and precision from 56% to 97.3% (§3.8), and the research-operations discipline — real-time decision log, run log, dry-run-first design — that made every number in this thesis traceable to the run that produced it (§3.2). The empirical contribution is narrower than originally planned and is stated as such: a systematically detected and increasingly accurate catalogue of cultural events (751 at the time of writing, §5.1), a transparent account of where the pipeline is and is not reliable (§5.3), and one closed, superseded structural finding about the account network kept for the record rather than advanced as a current result (§7.4.1). The applied contribution is the deployed bilingual discovery website itself (§4.5), with its full publication lifecycle — detection, staged human review, ranking, retention and manual deploy (§3.12) — documented as part of the method rather than left as background infrastructure.

7.2 Answering the Research Question

The thesis’s central question asked whether culturally-relevant activity in the Paris region could be systematically mapped, analysed and made discoverable from public Instagram data. The practical sub-question has the clearest answer: yes, to the extent that a live, bilingual, static website now surfaces a continuously refreshed, human-reviewed catalogue of such events, with its own retention policy and no dependency on a backend or user accounts (§4.5, §3.12). The thematic sub-question has a qualified answer: the pipeline

classifies detected events into the project’s priority domains with measured, improving accuracy (§3.8), but two of the four founding thematic pillars — photography and cinema — remain under-detected relative to their curated account base, a gap traced to the fixed sixteen-label taxonomy rather than to an absence of activity (§5.1). The structural sub-question — who are the connective actors in this network — has the least settled answer of the three: it was addressed once, early, before the scope broadened and before curation-based cleanup ran, and that answer is reported as a closed, dated finding rather than revalidated here (§7.4.1); answering it properly on the current, curated graph is left as future work rather than claimed prematurely.

7.3 Limitations

The limitations catalogued in §3.14 amount to five boundaries on how far these results generalise. The corpus is single-platform: organisations that announce primarily elsewhere are structurally invisible to it. It is geographically concentrated but not geographically clean: Île-de-France is the intended center of gravity, but institutions physically outside France enter through curation and through automatic discovery alike. The evaluation measures agreement with a single annotator’s judgement, not a consensus ground truth. The corpus’s boundary is set by how much one researcher can review by hand, which is the direct cost of replacing automated scoring with human curation (§3.9) — a trade this thesis argues was the right one on the evidence, but a real one. And every figure in this thesis is a snapshot of a graph and a pipeline that were both still changing as it was written, stated with its run of origin for exactly that reason.

7.4 Future Work

7.4.1 Revisiting Network Analysis on the Curated Graph

The account-network analysis described in §3.5 — exact PageRank and betweenness centrality, Leiden community detection at three resolutions, k-core decomposition, participation coefficient and E-I index, computed separately over a social layer (mentions, tags, co-authorship) and an algorithmic layer (Instagram’s own `relatedProfiles` suggestions) — was run once, at RUN-015 (2026-07-24), against 7,269 `:Account` nodes, before the scope broadened beyond the Latin American diaspora and before `cleanup_legacy_accounts.py` removed the uncurated periphery. At that snapshot the graph was 92.5% individual accounts against a small institutional core (cultural institutions 2.5%, state institutions 2.2%, commercial accounts 2.1%, media 0.7%); both layers were dominated by a single giant component (85–90% of nodes); Leiden modularity exceeded 0.85 at every resolution tested; and, most consequentially, some of the accounts

with the highest betweenness and PageRank on both layers had almost no posts of their own, their centrality derived entirely from being mentioned and tagged by others — the single most direct empirical argument, in this project’s own record, for why an account-selection criterion should not read relevance off an account’s own content.

That analysis was never repeated, and this thesis treats it as a closed, dated finding rather than a current result, for the reasons given in §3.5: the criterion it was built to serve was retired, and every precondition of the snapshot above — the corpus’s scope, its composition, and which accounts count as noise — has since changed. What remains genuinely open, and is left as future work rather than attempted here, is whether the same structural pattern (a small institutional bridging core amid a large organic periphery) survives on the graph as it stands today, restricted to the subgraph that human curation has actually validated rather than the full discovered periphery. That question is now cheaper to ask than it was at RUN-015 — the curated subgraph is a small fraction of the 7,269-node graph the first analysis ran over — and, unlike the original account-selection use case, would not be asking the network to do a job human curation already does better; it would be asking a documented, sociologically-motivated question about a graph the project already has.

7.4.2 Pipeline and Product

Beyond the network-analysis question above, the project’s own planning notes and open items point to a concrete list: a new account classifier with explicit geography and cultural-relevance dimensions, should account discovery ever need to scale beyond what manual curation can review; an expanded seed set from Latin American consulates in Paris (DD-022 lists the handles, several still unverified); a revised harvest cycle prioritising cheap profile scraping before expensive post scraping; a literature label added to the event taxonomy and a backfill decision for `eventArtTags` (DD-042), directly motivated by the under-detection of photography and cinema events noted in §5.1; resolution of the open question in DD-035 on ongoing exhibitions; normalisation of stylised Unicode digits in captions if confirmed frequent (DD-036); the remaining site screens — a clickable arrondissement and commune map, an explore view, and organiser-profile pages; a backfill decision for `titleFr/descriptionFr` on events created before DD-051, together with a first quality check of the translations it produces; whether DeepSeek should remain in the default cloud cascade at all, given the empty-response failures of DD-050; and collapsing data export, static page generation and deploy (§3.12) into a single command so publishing stops being the step that gets forgotten (DD-053).

7.5 Closing Reflection

[TODO: This one is deliberately left for Diego to write in the first person, not drafted here: a short, honest note on whether and how the project succeeded by its own fourth, most human objective (§1.3) — leading to actually meeting someone through it — separate from and not reducible to the technical metrics reported in Chapter 5. **]**

Appendix A

Graph Schema Reference

Table A.1: Node labels in the Hub Cultural graph.

Label	Origin	Notes
:Account	Ingestion	Also :Public or :Private; optionally :Political
:Post	Ingestion	Caption is the unit of event detection
:IgtvVideo	Ingestion	Treated like a post for authorship
:Comment	Ingestion	Author linked via WROTE
:Hashtag	Ingestion	
:Track	Ingestion	Audio track used in a post
:Location	Ingestion / events	Geocoded in phase 6
:Event	Event pipeline	Created only when the layer-3 gate passes; also :PendingReview until
:Arrondissement	Geocoding	Derived from postal codes 750XX
:City	Geocoding	
:Country	Geocoding	

Table A.2: Relationship types.

Group	Types
Ingestion	PUBLISHED, HAS_HASHTAG, MENTIONS, TAGS_USER, COAUTHORED_BY, TAGGED_AT, USES_MUSIC, WROTE, ON, LOCATED_AT, RELATED_TO
NLP and geo	MENTIONS_EVENT, ORGANIZED, PARTICIPATED_IN, SUPPORTED, LOCATED_IN

[TODO: Add an entity-relationship diagram and a property-level table for :Account and :Event, distinguishing ingestion-written properties from curation-written ones (artType, culturalIdentity, geoZone, manualFollowersCount, manualDataCuratedAt) and analysis-written ones (pageRankExact, betweennessExact, participationCoef, kCore, Leiden community labels). Source: docs/data_flow.md and the COLUMN_MAP of load_manual_account_categorization.py.]

Appendix B

Selected Decision Log Entries

[TODO: Translate and reproduce in full the entries most load-bearing for the argument: DD-006 (seed strategy), DD-013 (GDS to igraph), DD-018 (Nominatim), DD-026 (seed source versus cultural target), DD-031/032 (geographic misclassification), DD-034/037/039 (date extraction failures), DD-036 (the 495-post retuning), DD-040/041 (resolver redesign and geographic guardrail), DD-043/044 (export and site), DD-047 (the `:PendingReview` staging gate and its human-review interface), DD-050 (the DeepSeek empty-content failure and cloud-provider cascade reorder), DD-051 (bilingual event extraction), DD-052 (the `geoZone` review filter and its account-versus-event caveat), and DD-053 (the manual deploy step and the two-subdomain confusion it exposed). Note in a preamble that the log is written in Spanish and that these are the author’s translations.

]

Appendix C

Evaluation Protocol and Samples

[TODO: New appendix. Reproduce the classification criterion used for the blind human labelling (including the adjudication recorded in DD-035), the per-round sample sizes and dates, and the confusion counts behind the figures in §3.8. Source files: `data_processed/eval_100_report.md`, `data_processed/claude_labels.json`, and the per-batch `eval_*.csv` files. **]**

Appendix D

Discovery Website — Screenshots

[TODO: Home screen, filter bar, event detail panel — capture once the site has been reviewed in a real browser and the outstanding punch list is cleared. **]**

Appendix E

Repository Structure

[TODO: Reproduce the phase-by-phase script table from `CLAUDE.md`'s Commands section, including the naming convention (`1_harvest_`, `2_build_`, `3_analyze_`, `4_enrich_`, `5_export_`, DD-019) and the folder conventions: `old/` for superseded or environment-incompatible scripts, `testing/` for verification and diagnostic scripts. **]**

[TODO: Create `thesis/references.bib` and uncomment the two bibliography lines above. No bibliography exists in the repository yet; the only literature reference recorded anywhere is Vertovec (2009) in DD-020, and it needs verification. **]**